Software Requirements Specification

Agricultural remote-sensing research platform (working name to be decided)

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Status	Requirements gathering. Not yet approved for development

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1. Introduction

1.1 Purpose of this document

This document describes what the platform must do, for whom, and under which constraints. It is written so that a developer, a designer or a researcher can read it without further explanation. It does not prescribe implementation details except where a decision has already been taken.

1.2 Product summary

A no-code research workbench for agricultural remote sensing. A researcher selects one or more areas on a map, chooses satellite and climate variables and a time window, prepares and analyses the data, fits models and inspects their explanations, all from an interface instead of code. Every analysis is built from interchangeable blocks, every result shows where it came from and how certain it is, and an AI assistant helps interpret results and find supporting literature when the researcher asks.

1.3 Definitions

Term	Meaning
Area	A polygon, rectangle or uploaded geometry selected by the user
Block	One step of an analysis (for example "NDVI", "quality masking", "wavelet decomposition", "Random Forest")
Analysis chain	An ordered set of blocks applied to one or more areas
Workspace	A saved project containing areas, chains and results
Provenance	The record of data source, dates, processing and parameters behind a result
Guardrail	An automatic methodological check that warns the user about a likely error
Interpreter	The assistant role that explains the user's own results
Librarian	The assistant role that answers questions from scientific literature, with citations
EE	Google Earth Engine
SPEI	Standardized Precipitation-Evapotranspiration Index
QA	Quality assessment band supplied with satellite products

1.4 Priority scheme

Priority	Meaning
Must	Required for the first version (v1)
Should	Planned for v1 if time allows, otherwise v2
Could	Desirable, planned for a later version
Won't (now)	Explicitly out of scope for the current releases

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2. Product vision

2.1 Problem

Researchers who study agriculture with remote sensing currently face three obstacles.

1. **Access requires code.** The most capable tool, Google Earth Engine, is used through JavaScript or Python. We assume that many agricultural researchers do not code or code little; this assumption is to be verified (assumption A-1, section 10).
2. **Existing no-code tools stop at extraction.** Services such as NASA AppEEARS and Climate Engine deliver data or monitoring maps, but not the analysis that follows: decomposition, lag and drought-memory analysis, models and their explanations.
3. **Analysis results are easy to over-trust.** Our own work on olive-orchard NDVI showed that three equally accurate models explain the same data differently, and that common attribution methods over-weight correlated variables. Researchers have no simple way to see how reliable a result is.

2.2 Mission

Help agricultural researchers turn satellite and climate data into understanding they can trust, without writing code and without hidden assumptions.

2.3 Vision

Any researcher, anywhere, can ask a question about land and crops and get an answer they can explain, defend and reproduce.

2.4 Geographic and social focus

The platform serves agricultural research in climate-stressed and under-monitored regions, starting with the Mediterranean basin and North Africa, where drought and heat are already affecting food security and where monitoring tools are fewer than in North America and northern Europe. Nothing in the design should prevent use elsewhere.

2.5 Principles

These principles decide trade-offs whenever requirements conflict.

1. **Trust first.** Every number shows where it came from, how it was processed and how uncertain it is.
2. **The researcher is in control.** This is a platform with AI, not an AI platform. The user builds the analysis; the assistant helps only when asked.
3. **Transparent by default.** Every result can show the code that produced it.
4. **Built for agriculture.** Depth in one domain rather than breadth across all of them.

5. **Calm surface, full depth.** A quiet, focused interface that still exposes every parameter to those who want it.

2.6 Positioning

Tool	What it does	What it lacks for this audience
Google Earth Engine Code Editor	Full analysis over a vast catalogue	Requires programming
NASA AppEEARS	Extracts satellite data for areas without code	No analysis, models or interpretation
Climate Engine	No-code maps and time series for climate and monitoring	Monitoring focus; no models, explanation or assistant
This platform	No-code analysis from data to models and explanations, with provenance, uncertainty and an assistant	

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3. Users

3.1 Primary user: agricultural researcher

Works in a university, research institute or agricultural agency. Understands agronomy and has basic remote-sensing knowledge. Codes little or not at all.

- **Goals:** characterise vegetation dynamics of crops and regions; relate them to climate and drought; compare areas and periods; produce defensible results for publication.
- **Pains:** time lost writing and debugging code; uncertainty about whether a method was applied correctly; difficulty explaining machine-learning results.

3.2 Secondary users

- **Graduate students** learning the field, who benefit from the tutor role and from seeing the code behind each result.
- **Researchers who code**, who use the platform for speed and export the code to continue in their own environment.

3.3 Not targeted now

Farmers, agribusinesses and consumers. A simplified product for them is a separate, later project (section 9).

3.4 Core user journey (first ten minutes)

1. The researcher opens the platform and sees a map.
2. They select a place by searching, drawing one or more areas, or uploading a file.
3. They choose the variables they want, for example NDVI, land surface temperature and SPEI.
4. They set the time window.
5. Charts appear for each area, with the source and processing of every series shown alongside.
6. They add blocks (decomposition, comparison, model) and inspect the results.

7. If something is unclear, they ask the assistant about the result in front of them.

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4. Functional requirements

4.1 Accounts and workspaces (ACC)

ID	Requirement	Priority
ACC-01	The user can sign in with a Google account.	Must
ACC-02	The user can sign in with an institutional account (university single sign-on).	Could
ACC-03	The user can sign in with ORCID.	Could
ACC-04	The user can connect their own Google Earth Engine project, and the platform uses it for all data requests made on their behalf.	Must
ACC-05	A user without an Earth Engine account can explore a demo mode built on pre-computed sample areas.	Should
ACC-06	The user can create, rename, save, reopen and delete workspaces.	Must
ACC-07	A workspace stores its areas, analysis chains, parameters and results.	Must

4.2 Map and areas (MAP)

ID	Requirement	Priority
MAP-01	The map supports search by place name and coordinates.	Must
MAP-02	The user can draw polygons and rectangles.	Must
MAP-03	The user can create several areas in one workspace and name each one.	Must
MAP-04	The user can upload areas as GeoJSON, KML or zipped shapefile.	Must
MAP-05	The user can switch basemaps: satellite imagery, terrain, street map and a dark minimal map.	Must
MAP-06	Drawing tools snap, show area size while drawing and allow vertex editing.	Should
MAP-07	The user can group areas (for example "irrigated orchards") and analyse a group as one unit or as members.	Should
MAP-08	The user can select areas from administrative boundaries or land-cover classes.	Could
MAP-09	The map displays the selected variable as a layer for a chosen date.	Should

4.3 Data (DATA)

ID	Requirement	Priority
DATA-01	The user chooses variables from a catalogue grouped by theme (vegetation, temperature, water and drought, climate, soil and terrain).	Must
DATA-02	The v1 catalogue contains at least MODIS NDVI, MODIS land surface temperature and SPEI at user-selectable accumulation periods (section 6).	Must
DATA-03	Values are aggregated over each area as an area-weighted mean, with the spatial standard deviation and pixel count reported per date.	Must

ID	Requirement	Priority
DATA-04	Each variable shows its source product, version, spatial resolution, temporal resolution and licence.	Must
DATA-05	The user can upload their own tabular data (for example yields or field measurements) linked to an area and dates.	Could
DATA-06	The catalogue is extended with further variables listed in section 6.	Should / Could
DATA-07	Data requests that cannot be served (area too large, missing coverage) produce a clear message stating the reason and the limit.	Must

4.4 Time (TIME)

ID	Requirement	Priority
TIME-01	The user sets a start and end date for the analysis.	Must
TIME-02	The user chooses the temporal resolution of results: native, monthly or seasonal.	Must
TIME-03	The user can define named periods (for example "drought 2022–2024") and compare them.	Should
TIME-04	The user can restrict analysis to months or growing seasons.	Should

4.5 Preparation (PREP)

ID	Requirement	Priority
PREP-01	Preparation steps are chosen by the user. Recommended steps are selected by default and each shows a one-line reason.	Must
PREP-02	Quality masking using the product's QA band, with the number of observations removed reported.	Must
PREP-03	Gap filling by interpolation, with the filled dates marked.	Must
PREP-04	Smoothing (for example Savitzky-Golay), optional.	Should
PREP-05	Decomposition into trend and seasonal components, with a choice of method: none, wavelet (discrete Meyer and others, level chosen by target period), STL, and others listed in section 6.	Should (wavelet), Could (others)
PREP-06	Anomalies relative to a user-chosen reference period.	Should
PREP-07	No preparation step is applied without being visible in the analysis chain.	Must

4.6 Analysis (ANA)

ID	Requirement	Priority
ANA-01	Descriptive statistics and seasonal climatology per area.	Must
ANA-02	Comparison of the same variable across areas and periods on one chart.	Must
ANA-03	Correlation and lag analysis between variables, computed on deseasonalised series by default.	Should
ANA-04	Trend testing (Mann-Kendall, Sen's slope) and breakpoint detection.	Could
ANA-05	Phenology metrics: start, peak and end of season, season length and amplitude.	Could
ANA-06	Drought-memory analysis relating vegetation to drought indices over several accumulation periods.	Could

ID	Requirement	Priority
ANA-07	Pixel-level maps of trends or anomalies within an area.	Could

4.7 Models (MOD)

ID	Requirement	Priority
MOD-01	The user can fit machine-learning models, starting with Random Forest, XGBoost and Ridge regression, to forecast a chosen variable from chosen predictors and lags. The model catalogue is extensible (for example linear models, support vector regression, gradient-boosting variants).	Should
MOD-02	Training and test periods are split by time, never at random; the default leaves the most recent period for testing.	Should
MOD-03	Results report RMSE, R^2 and MAE against persistence and climatology baselines.	Should
MOD-04	Results report a noise floor from repeated fits with different random seeds.	Should
MOD-05	Hyperparameters can be left at tuned defaults or set manually.	Should
MOD-06	Deep-learning models for time series (for example LSTM, GRU, temporal convolutional networks, transformers).	Could

4.8 Explanations and trust (XAI)

ID	Requirement	Priority
XAI-01	Each fitted model can be explained with SHAP and permutation importance.	Should
XAI-02	Drop-column importance is available for confirming whether a model needs a feature.	Could
XAI-03	When several models are fitted, their explanations are compared and the level of agreement is shown.	Could
XAI-04	Importances are shown with their variation across seeds, not as single numbers.	Should

4.9 Visualisation (VIS)

ID	Requirement	Priority
VIS-01	Interactive time-series charts with zoom, hover values and toggling of series.	Must
VIS-02	Overlaid comparison of several areas or periods.	Must
VIS-03	Seasonal-cycle charts (values by month across years).	Must
VIS-04	Uncertainty is drawn where it exists: spatial spread, seed variation, confidence intervals.	Must
VIS-05	Every chart has a provenance panel showing sources, dates and processing.	Must
VIS-06	Charts can be downloaded as image files and their data as CSV.	Should
VIS-07	A focus mode hides everything except the current element: a chart, the map or an analysis panel.	Should

4.10 AI assistant (AI)

The assistant is optional and appears only when the user opens it.

ID	Requirement	Priority
AI-01	**Interpreter.** Answers questions about the user's current results, using the computed numbers and provenance as its source, and refers to the specific chart or value it relies on.	Should
AI-02	**Librarian.** Answers questions from a curated corpus of agricultural and remote-sensing literature, and every answer cites its sources with links.	Could
AI-03	**Guardrail.** Methodological checks run as deterministic code (for example: correlating two seasonal series, predictors too correlated to attribute, test period overlapping training). The assistant only explains a triggered check and cites the reference behind it.	Should
AI-04	**Tutor.** Explains concepts (for example what SPEI-24 means) at the user's level.	Could
AI-05	The assistant declines questions outside agriculture, remote sensing and research methods.	Must (when the assistant exists)
AI-06	The assistant states when it does not know, and never invents numbers or references.	Must (when the assistant exists)
AI-07	The assistant can suggest blocks, but never adds or runs them without the user's confirmation.	Must (when the assistant exists)
AI-08	**Team of agents.** The assistant is organised as several specialised agents that the user can call individually or together, like a research team: for example a librarian, a builder that proposes analysis chains, an interpreter and a methods reviewer (guardrail). A farmer-oriented agent belongs to the later farmer product.	Could
AI-09	**Closed-access literature.** When a relevant paper is not open access, the librarian lists it with its reference and how it can be obtained, without using its content.	Could
AI-10	**Personal library.** The user can upload their own papers, which the librarian then uses for that user only.	Could

4.11 Reproducibility (REP)

ID	Requirement	Priority
REP-01	Every analysis chain can be exported as an executable Python notebook that reproduces its results.	Should
REP-02	The same chain with the same inputs always gives the same results (fixed random seeds, versioned data products).	Must
REP-03	A workspace records the platform version and data-product versions used.	Must

4.12 Sharing (SHR)

ID	Requirement	Priority
SHR-01	The owner can share a workspace with colleagues as view-only or editable.	Could
SHR-02	Comments can be attached to charts.	Could

4.13 Export (EXP)

ID	Requirement	Priority
EXP-01	A summary of a workspace (charts, parameters, provenance) can be exported as a document. Exporting is secondary to exploring and discussing results.	Could

5. Non-functional requirements

5.1 Trust and correctness (NFR-T)

ID	Requirement
NFR-T1	Each analysis block is covered by automated tests.
NFR-T2	Each analysis block is validated against an independent reference implementation or published results, within stated numerical tolerances, before release.
NFR-T3	Default settings follow documented best practice, and the reference for each default is recorded.
NFR-T4	Every methodological guardrail is backed by a cited source.

5.2 Performance (NFR-P)

ID	Requirement
NFR-P1	Interface actions respond within one second.
NFR-P2	Data requests and computations longer than a few seconds run as background jobs, with progress shown and the interface remaining usable.
NFR-P3	Results already computed for the same inputs are reused rather than recomputed.
NFR-P4	Target times and maximum area sizes are set after load testing (open question O-6).

5.3 Usability and design (NFR-U)

ID	Requirement
NFR-U1	Visual direction: calm, dark and minimal (reference: Endel), with technical depth and full control of parameters available (reference: Ableton).
NFR-U2	Dark theme by default, with a light theme available.
NFR-U3	Text and charts meet WCAG AA contrast.
NFR-U4	Advanced parameters are hidden behind an expandable section, never removed.
NFR-U5	A first-time user completes the core journey (section 3.4) without documentation.

5.4 Security and privacy (NFR-S)

ID	Requirement
NFR-S1	Authentication uses standard protocols (OAuth 2.0 / OpenID Connect).
NFR-S2	Earth Engine credentials are stored encrypted and used only for the owner's requests.
NFR-S3	Workspaces and uploaded data are private unless the owner shares them.
NFR-S4	Users can delete their account and all associated data.

5.5 Portability and deployment (NFR-D)

ID	Requirement
NFR-D1	The platform runs as a web application in the cloud.
NFR-D2	The platform can also be installed and run on a researcher's own computer.
NFR-D4	All functionality is available through a documented API, so the platform can be used programmatically as well as through the interface. The API may be built first (open question O-10).
NFR-D3	All components are containerised.

5.6 Maintainability (NFR-M)

ID	Requirement
NFR-M1	The analysis engine is a separate package with a documented interface, used by the interface and by exported notebooks alike.
NFR-M2	New blocks can be added without changing existing ones.
NFR-M3	The data layer is abstracted so that sources other than Earth Engine can be added.

5.7 Licensing and compliance (NFR-L)

ID	Requirement
NFR-L1	Earth Engine is used under its terms; non-commercial use requires each user's own registered project.
NFR-L2	Each data product's licence and required citation are displayed and included in exports.
NFR-L3	Literature in the librarian corpus is used only within its licence. Open-access papers may be indexed; closed-access papers are only referenced (AI-09) unless the user supplies them (AI-10).

6. Data catalogue

Theme	Variable	Product	Resolution	Cadence	Priority
Vegetation	NDVI, EVI	MODIS MOD13Q1	250 m	16-day	Must
Vegetation	NDVI and other indices	Sentinel-2	10–20 m	~5-day	Should
Vegetation	NDVI and other indices	Landsat 8/9	30 m	16-day	Could
Temperature	Land surface temperature, day and night	MODIS MOD11A2	1 km	8-day	Must
Drought	SPEI, 1–48 months	SPEIbase (CSIC)	0.5°	monthly	Must
Water	Precipitation	CHIRPS	~5 km	daily	Should
Climate	Air temperature, soil moisture, vapour-pressure deficit	ERA5-Land	~9 km	hourly/monthly	Should
Water	Evapotranspiration	MODIS MOD16	500 m	8-day	Could
Radar	Backscatter (all-weather)	Sentinel-1	10 m	~6-day	Could
Land cover	Land-cover class	ESA WorldCover / Dynamic World	10 m	annual / near-real-time	Could

Theme	Variable	Product	Resolution	Cadence	Priority
Soil and terrain	Soil properties, elevation, slope	SoilGrids, SRTM	250 m, 30 m	static	Could

The cadence column gives each product's native time step. The user can analyse data at the native step or aggregate it to any coarser step (for example monthly or seasonal); a finer step than the product provides is not possible.

Decomposition methods: none; wavelet (Should); STL, BFAST, EMD (Could).

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7. System overview

```
Browser or desktop app (React, interactive map, charts)
            │
            ▼
API service (Python, FastAPI) ─── Authentication (OAuth: Google, institutional, ORCID)
            │
            ├── Job queue and workers ── Analysis engine (Python package: preparation,
            │                            decomposition, analysis, models, explanations)
            │
            ├── Data layer ── Google Earth Engine (user's project) ; later other catalogues
            │
            ├── Database (workspaces, areas, provenance) ; result cache ; file storage
            │
            └── Assistant service ── language model ; literature index ; guardrail rules
```

Decisions taken: Python analysis engine as a separate package; React web interface with an interactive map; background jobs for long computations; containerised deployment; Earth Engine accessed through each user's own project.

Decisions open: cloud provider (Google Cloud, which hosts Earth Engine, or AWS); database and cache products; language-model provider; desktop packaging; open-source or proprietary licence.

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8. Release plan

8.1 Version 1 (target: one week)

A working end-to-end path on the researcher's own computer:

- Sign in with Google and connect an Earth Engine project (ACC-01, ACC-04)
- Map with search, basemaps, several drawn or uploaded areas (MAP-01 to MAP-05)
- NDVI, LST and SPEI with quality masking and gap filling (DATA-01 to DATA-04, PREP-01 to PREP-03, PREP-07)
- Time window and resolution (TIME-01, TIME-02)
- Time-series, comparison and seasonal charts with uncertainty and provenance (VIS-01 to VIS-05, ANA-01, ANA-02)
- Workspaces saved locally (ACC-06, ACC-07)

Acceptance: for a reference area, the extracted NDVI, LST and SPEI series match an independent extraction within stated tolerances.

8.2 Version 2

Decomposition, correlation and lag analysis, models with explanations and noise floors, interpreter and guardrails, code export, cloud deployment.

8.3 Later

Librarian, tutor, further data products, phenology, sharing, institutional sign-in, user data uploads, and a separate product for farmers and agribusiness.

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9. Out of scope (now)

- A simplified product for farmers, agribusinesses and consumers
- Real-time alerting and operational monitoring
- Commercial use of Earth Engine through a shared platform account
- Mobile applications

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10. Open questions for the client

ID	Question
O-1	Which variables do you and your students use most, beyond NDVI, LST and SPEI?
O-2	Which analyses take the most time today, and which would you most like to do without code?
O-3	Which crops and regions should the first version be tested on?
O-4	What area sizes are typical: single fields, orchards, districts or whole governorates?
O-5	Do your projects need user-supplied data such as yields or field measurements?
O-6	What waiting time is acceptable for a large analysis?
O-7	Would your lab test early versions and give feedback?
O-8	Which sources should the librarian's literature corpus include first?
O-9	Do the researchers you work with code? (verifies assumption A-1)
O-10	Should the first version be an API, a web application or both?

Assumptions to verify

ID	Assumption	How to verify
A-1	Many agricultural researchers do not code or code little.	Literature on researchers' tool use; short survey of the client's lab and contacts

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11. Risks

Risk	Effect	Mitigation
Earth Engine quotas or terms change	Data requests fail or become costly	Abstract the data layer (NFR-M3); cache results

Risk	Effect	Mitigation
The assistant states incorrect facts	Loss of trust	Ground answers in computed results and cited literature; deterministic guardrails; refusal outside the domain
Scope grows faster than development	No usable version	Strict release plan (section 8); priorities enforced
Analysis errors in the engine	Wrong scientific conclusions	Reference-result acceptance tests (NFR-T2); automated tests
Large areas are slow	Poor experience	Background jobs, caching, stated limits